Looking Away

Building 2: Loops

Ways between ways, and the systems that come back.

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A later chapter builds a system and blurs it by the action of an earlier one; a *loop* is the theorem saying the earlier system comes back (draft rule F8). Building 2 also holds the levels above moves: ways between ways (Chapter 16 is its first room).

Chapter numbers restart in this book; Building 0 and Building 1 are cited by name.

Planned, not yet written (`plans/TERRITORIES.md`, Leg 7). Loops; Amplify (draft rule Z12); Atoms; the missing move.

Part I. Words about words

Building 0, Chapter 0 writes rules about words; Building 1, Chapter 2 builds words and glues them. Here the words are statements and proofs, and the blurs are "forget the proof" and "forget everything but the truth".

Chapter 1. Proofs

statements → what every world says · from: Building 0, Chapters 0, 3; Building 1, Chapter 2 · build, blur

Statements and proofs are words. Glue two statements when each proves the other; glue them when every world gives them the same answer. The two blurs agree: that is completeness. For the single world of the whole numbers they do not: that is incompleteness.

Written for statements built from letters with "not", "and", "or", "if". The lines for statements with "for all" are stated and reported (T4, T5); their proofs need a design of their own.

Given

G1 (Building 0, Chapter 0, M3.) Words in a list of names; maps on words by recursion.

G2 (Building 1, Chapter 2.) Words, glued by rules from below; B2 by an action on words.

G3 (Building 0, Chapter 3.) $\{0,1\}$ as $\mathbb{Z}/2$.

Definitions

D1 (Build, M3.) *Statements*: words in letters $p_1, p_2, \ldots$, the name $\bot$, and $\neg$, $\wedge$, $\vee$, $\to$. Write $A \leftrightarrow B$ for $(A \to B) \wedge (B \to A)$.

D2 (Build.) A *proof* of A: a list of statements ending in A, each one an instance of $A \to (B \to A)$, $(A \to (B \to C)) \to ((A \to B) \to (A \to C))$, $(\neg B \to \neg A) \to (A \to B)$ (with $\neg A := A \to \bot$ and $\wedge$, $\vee$ written through $\to$, $\bot$), or got from two earlier ones B and $B \to A$. Write $\vdash A$. Checking a proof is a recursion; finding one is a search.

D3 A *world*: an arrow from the letters to $\{0,1\}$. It gives every statement a value, by recursion ($\bot \mapsto 0$, and the usual tables).

The blur

On statements in the letters $p_1, \ldots, p_n$. O: $\neg$, $\wedge$, $\vee$, $\to$.

B1 (From below: forget the proof.) $A \sim B$ when $\vdash A \leftrightarrow B$.

B2 (From above: forget all but the answers.) Questions: the value of a statement in each world.

Propositions

P1 (Survives.) $\neg$, $\wedge$, $\vee$, $\to$ all respect B1 and B2. The bags of B2 form $\{0,1\}^{(\text{worlds})}$, with the operations taken world by world.

P2 (Dies: length; the proof.) p and $\neg\neg p$ are glued, of lengths 1 and 3. "Which proof" is the first thing forgotten; "how short a proof can be" dies with it, and is a question of its own (R2).

P3 (Normal form.) Each bag of B2 holds exactly one "or" of full rows, one row for each world giving 1 (in a fixed order). So there are 2^{2^n} bags.

Theorems

T1 (One way: sound.) If $A \sim B$ then A and B agree in every world. (Each axiom is 1 in every world, and "from B, $B \to A$ get A" keeps "1 in every world".)

T2 (The other way. A construction.) If A is 1 in every world, $\vdash A$. (For each world w, prove "the row of $w \to A$" by recursion on A; then remove the letters one at a time with $\vdash (p \to A) \to ((\neg p \to A) \to A)$.)

T3 (Appears: completeness.) B1 and B2 agree. So the bags of B1 are exactly 2^{2^n}, with P3 as normal forms, and "provable" is "true in every world".

T4 (Reported, classical: with "for all".) For statements about a floor with "for all" and "there is", and a list of axioms Γ: B1 (each proves the other from Γ) and B2 (same value in every floor obeying Γ) agree. The construction in the proof builds a floor out of the words themselves.

T5 (Reported, classical: Appears fails for one world.) Take the axioms of $\mathbb{N}$ (Building 0, Chapter 1, with $+$, $\times$) and replace B2 by the single world $\mathbb{N}$. Then B1 is strictly finer: some statement G is true in $\mathbb{N}$ and is not glued to $\top$. Z4 holds, Z5 fails (Building 0, Chapter 0), with G as witness. The proof codes words as numbers, so that statements about $\mathbb{N}$ speak about statements: a loop of the kind this building is named for.

Remarks

R1 (A twist.) A statement of the form "for every n, a check on n comes out yes" can be refuted by one finite witness if it is false. So if it is shown to be neither provable nor refutable, it is true in $\mathbb{N}$. Many open questions about primes (Building 0, Chapter 18) have this form.

R2 (How much proof.) Two sizes, both what dies in B1: *length*, the shortest proof, and *strength*, the first rung of a ladder of axiom lists that proves A. The pigeonhole statement for $n+1$ birds needs proofs of length growing faster than any power of n in the weakest systems (classical, reported), and only a few lines once counting bags is allowed (Building 1, Chapter II.6 uses counts this way).

R3 (Old names.) D1: propositional formulas; D2: a Hilbert system; D3: valuations; B1: the Lindenbaum algebra, a Boolean algebra; T1: soundness; T2, T3: completeness (Post; Kalmár's proof); T4: Gödel's completeness theorem; T5: Gödel's first incompleteness theorem.

R4 (Machine check, `checks/b2-ch1-2-check.py`.) The axioms on many instances are 1 in every world; $2^{2^2} = 16$ bags for two letters; P3 normal forms.

Exercises

E1 Write the normal form of P3 for $p \to q$ and for $\neg(p \leftrightarrow q)$.

E2 Show $\vdash p \to p$ from the axioms of D2 in five lines.

E3 Why does T2 not give a fast way to find proofs?

Chapter 2. Equally provable

statements → how provable they are · from: Chapter 1; Building 0, Chapters 18, 19, 34
· blur, ladder

Glue two statements when the theory proves "this one is provable exactly when that one is". Without letters the bags are a ladder of shape $\omega + 1$, and a statement's rung is the first depth at which it fails: a valuation, like v_p.

All the lines about arithmetic here rest on three classical results, reported: the conditions on $\Box$ (D1), Löb's theorem (G2), and the depth reading (G3). The chapter's own work is the blur and its reading.

Given

G1 (Chapter 1, T5.) Statements of arithmetic can speak of proofs. For a statement A, $\Box A$ is the statement "A is provable", and $\vdash A$ gives $\vdash \Box A$; $\vdash \Box(A \to B) \to (\Box A \to \Box B)$; $\vdash \Box A \to \Box\Box A$ (reported, classical).

G2 (Reported, classical: Löb.) $\vdash \Box(\Box A \to A) \to \Box A$.

G3 (Reported, classical: depth.) For statements built from $\bot$ with $\neg$, $\wedge$, $\vee$, $\to$, $\Box$ (no letters): read A at each depth $n = 0, 1, 2, \ldots$, with $\Box A$ true at depth n when A is true at every depth below n. Then $\vdash A$ exactly when A is true at every depth.

The blur

On statements of arithmetic. O: $\wedge$, $\Box$.

B1 (From below.) $A \approx B$ when $\vdash \Box A \leftrightarrow \Box B$: "equally provable".

B2 (From above, without letters.) Question: $\rho(A) :=$ the first depth at which A is false (∞ if none), the *rung* of A.

Propositions

P1 (Survives.) $\wedge$ and $\Box$. ($\Box(A \wedge B) \leftrightarrow \Box A \wedge \Box B$ by G1.)

P2 (Appears inside: Löb.) $A \approx (\Box A \to A)$ for every A. (G1 one way, G2 the other.)

P3 (Dies: truth.) $\bot \approx \neg\Box\bot$ (P2 with $A = \bot$). $\neg\Box\bot$, "consistency", is true, and $\bot$ is false.

P4 (Dies: $\vee$ and $\neg$.) $\bot \approx \neg\Box\bot$, but $\bot \vee \Box\bot$ is not provable while $\neg\Box\bot \vee \Box\bot$ is.

Theorems

T1 (Appears, without letters: the bags are the rungs.) $A \approx B \iff \rho(A) = \rho(B)$. ($\Box A$ is true at depth n exactly when $n \leq \rho(A)$; G3.)

T2 (A ladder of shape $\omega + 1$.) Every rung $k = 0, 1, 2, \ldots$ and ∞ occurs: $\rho(\Box^k\bot) = k$, and the provable statements are rung ∞. For every A with $\rho(A) = k < \infty$:

$\vdash \Box A \leftrightarrow \Box^{k+1}\bot$, so $\Box^k\bot$ is a normal form of its bag. No two rungs are glued: that would give $\vdash \Box^{k+2}\bot \to \Box^{k+1}\bot$, and Löb would give $\vdash \Box^{k+1}\bot$.

T3 (Survives, read on the rungs.) $\rho(\Box A) = \rho(A) + 1$; $\rho(A \wedge B) = \min(\rho(A), \rho(B))$. These are the laws of v_p (Building 0, Chapter 18, D4) and of ord_a (Building 0, Chapter 34, P1) with "$\times$" read as "$\wedge$": a rung is the first place a statement stops being glued to the provable ones, as $v_p(a)$ is the first rung at which a stops being glued to 0.

T4 (Placement.) $\bot$ and consistency $\neg\Box\bot$ on rung 0; $\Box\bot$ ("inconsistent") on rung 1; "$\Box\bot \to \Box\Box\bot$" on rung ∞.

Remarks

R1 (What is known.) T1–T3 are the letterless normal form of the logic of $\Box$ (Boolos, classical). Adding stronger boxes stretches the ladder to the ordinal ε_0, the measure of strength of arithmetic (Beklemishev, reported). The blur B1 with letters, and the mixture of B1 with "same consequences of the form for every n", are not worked out: open.

R2 (The twin.) The rung is a valuation: it is the fourth time the book meets "first rung at which not glued" (Building 0, Chapters 19, 21, 34). Here the ladder stops at $\omega + 1$ instead of having a bottom.

R3 (Old names.) $\Box$: the provability predicate; G1: the Hilbert–Bernays–Löb conditions; G2: Löb's theorem; G3: Solovay's completeness for GL on letterless sentences, read on a line of worlds; P3: Gödel's second incompleteness theorem (consistency is not provable) seen as "truth dies".

R4 (Machine check, `checks/b2-ch1-2-check.py`.) On letterless statements up to size 3 read at depths below 40: T1 (bags are rungs), P2 (Löb instances), T3 ($\Box$ climbs one rung, $\wedge$ is min), T4, P4.

Exercises

E1 Find the rung of $\Box\bot \to \bot$ and of $\Box\Box\bot \wedge \neg\Box\bot$.

E2 Why is $\vee$ not max on rungs? Find $\rho(A \vee B)$ for $A = \bot$, $B = \Box\bot$.

E3 Show from P2 that $A \approx \Box A \to A \approx \Box(\Box A \to A) \to (\Box A \to A)$.